Ellipses and Quadratic Forms

0.1 Ellipse centered at the origin

Let us begin by recalling the formula for an ellipse centered at the origin who's major principle axis is aligned with the x-axis and minor principal axis is aligned with the y-axis:

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

where a is the semi-major axis and b is the semi-minor axis.

If we wish to rotate the axes then we use the rotation matrix:

$$\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

to transform the coordinates (x, y) to new coordinates (x', y'):

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

This gives us:

$$x' = x\cos\theta - y\sin\theta$$

$$y' = x\sin\theta + y\cos\theta$$

And the equation for the ellipse in the rotated coordinate system becomes:

$$\frac{(x\cos\theta - y\sin\theta)^2}{a^2} + \frac{(x\sin\theta + y\cos\theta)^2}{b^2} = 1$$

In particular the equation for an ellipse with semi-major axis along the ine y=x and semi-minor axis along the line y = -x is given by setting $\theta = \frac{\pi}{4}$, and we get

$$\frac{(x-y)^2}{2a^2} + \frac{(x+y)^2}{2b^2} = 1$$

0.2 Positive Definite Matrices

Theorem 1. *An ellipse can also be represented by a quadratic form:*

$$\begin{bmatrix} x & y \end{bmatrix} \begin{bmatrix} A & B \\ B & C \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 1$$

where the matrix$M = \begin{bmatrix} A & B \\ B & C \end{bmatrix}$ *is positive definite (symmetric with non-negative eigenvalues).*

Proof. When we diagonalize the matrix M using its eigenvectors, we obtain the spectral decomposition:

$$M = QDQ^T$$

where Q is an orthogonal matrix (rotation matrix) whose columns are the eigenvectors of M, and D is a diagonal matrix containing the eigenvalues λ_1, λ_2.

The quadratic form can now be rewritten as:

$$\mathbf{v}^T M \mathbf{v} = \mathbf{v}^T Q D Q^T \mathbf{v} = (Q^T \mathbf{v})^T D (Q^T \mathbf{v}) = 1$$

where

$$\mathbf{v} = \begin{bmatrix} x \\ y \end{bmatrix}$$

. Let $\mathbf{v}' = Q^T \mathbf{v}$ represent the coordinates in the rotated coordinate system where

$$\mathbf{v}' = \begin{bmatrix} x' \\ y' \end{bmatrix}$$

Then:

$$\mathbf{x}'^T D \mathbf{x}' = \lambda_1 x'^2 + \lambda_2 y'^2 = 1$$

Re-arranging we get:

$$\frac{x'^2}{1/\lambda_1} + \frac{y'^2}{1/\lambda_2} = 1$$

This is the standard equation of an ellipse with semi-axis lengths:

$$a = \frac{1}{\sqrt{\lambda_{\min}}}, \quad b = \frac{1}{\sqrt{\lambda_{\max}}}$$

where $\lambda_{\min}$ is the smaller eigenvalue (corresponding to the major axis) and $\lambda_{\max}$ is the larger eigenvalue (corresponding to the minor axis).

Therefore, the semi-axis lengths are given by:

$$\boxed{a = \frac{1}{\sqrt{\lambda_1}}, \quad b = \frac{1}{\sqrt{\lambda_2}}}$$

where $\lambda_1 \leq \lambda_2$ for the standard convention with $a \geq b$. □

The following lemma will be useful later.

Lemma 1. *Let* $M = \begin{bmatrix} 1 & \rho \\ \rho & 1 \end{bmatrix}$ *be a correlation matrix with* $|\rho| < 1$. *Then:*

1. *The eigenvalues of* M *are* $\lambda_1 = 1 + \rho$ *and* $\lambda_2 = 1 - \rho$.
2. *Corresponding eigenvectors are* $\mathbf{v}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ *and* $\mathbf{v}_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$.
3. *The quadratic form* $\mathbf{x}^T M^{-1} \mathbf{x} = 1$ *defines an ellipse centered at the origin with:*
4. *Semi-axis length* $\sqrt{1+\rho}$ *along the direction of* $\mathbf{v}_1$ *(line* $y = x$*)*
5. *Semi-axis length* $\sqrt{1-\rho}$ *along the direction of* $\mathbf{v}_2$ *(line* $y = -x$*)*

Proof. We prove each part of the theorem sequentially.

Part 1: Eigenvalues

The eigenvalues λ satisfy the characteristic equation $\det(M - \lambda I) = 0$:

$$\det \begin{bmatrix} 1-\lambda & \rho \\ \rho & 1-\lambda \end{bmatrix} = (1-\lambda)^2 - \rho^2 = 0$$

Expanding and solving:

$$(1-\lambda)^2 = \rho^2 \implies 1 - \lambda = \pm\rho \implies \lambda = 1 \mp \rho$$

Therefore, the eigenvalues are:

$$\lambda_1 = 1 + \rho, \quad \lambda_2 = 1 - \rho$$

Since $|\rho| < 1$, both eigenvalues are positive, confirming that M is positive definite.

Part 2: Eigenvectors

For $\lambda_1 = 1 + \rho$, solve $(M - \lambda_1 I)\mathbf{v} = \mathbf{0}$:

$$\begin{bmatrix} 1-(1+\rho) & \rho \\ \rho & 1-(1+\rho) \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} -\rho & \rho \\ \rho & -\rho \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

This yields the equation $-\rho v_1 + \rho v_2 = 0$, which simplifies to $v_1 = v_2$. Choosing $v_1 = 1$, we obtain the eigenvector:

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

For $\lambda_2 = 1 - \rho$, solve $(M - \lambda_2 I)\mathbf{v} = \mathbf{0}$:

$$\begin{bmatrix} 1-(1-\rho) & \rho \\ \rho & 1-(1-\rho) \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} \rho & \rho \\ \rho & \rho \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

This yields the equation $\rho v_1 + \rho v_2 = 0$, which simplifies to $v_1 = -v_2$. Choosing $v_1 = 1$, we obtain the eigenvector:

$$\mathbf{v}_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

Note that $\mathbf{v}_1$ and $\mathbf{v}_2$ are orthogonal: $\mathbf{v}_1^T \mathbf{v}_2 = 1 \cdot 1 + 1 \cdot (-1) = 0$.

This is the standard form of an ellipse with semi-axis lengths $\sqrt{1+\rho}$ and $\sqrt{1-\rho}$.

Since the columns of Q are the normalized eigenvectors $\frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and $\frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$, the principal axes align with the lines $y = x$ and $y = -x$.

For $\rho \geq 0$, $\sqrt{1+\rho} \geq \sqrt{1-\rho}$, so:

Semi-major axis length $= \sqrt{1+\rho}$ along $y = x$

Semi-minor axis length $= \sqrt{1-\rho}$ along $y = -x$

This completes the proof. □